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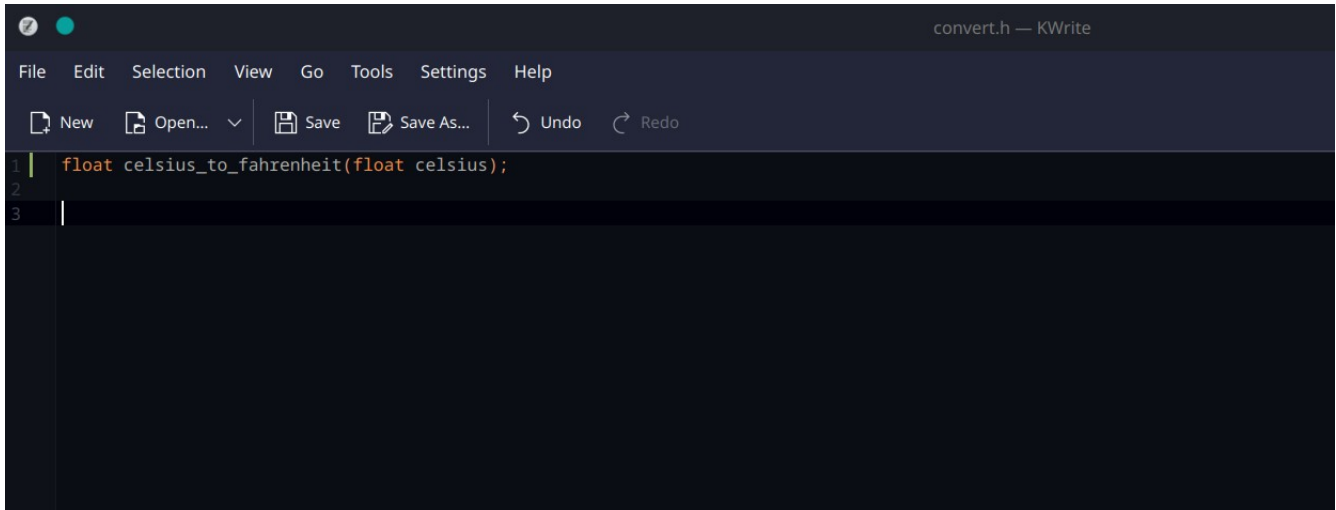
**Operating System Lab**

**Remaining Lab Task 2**

## Task :2

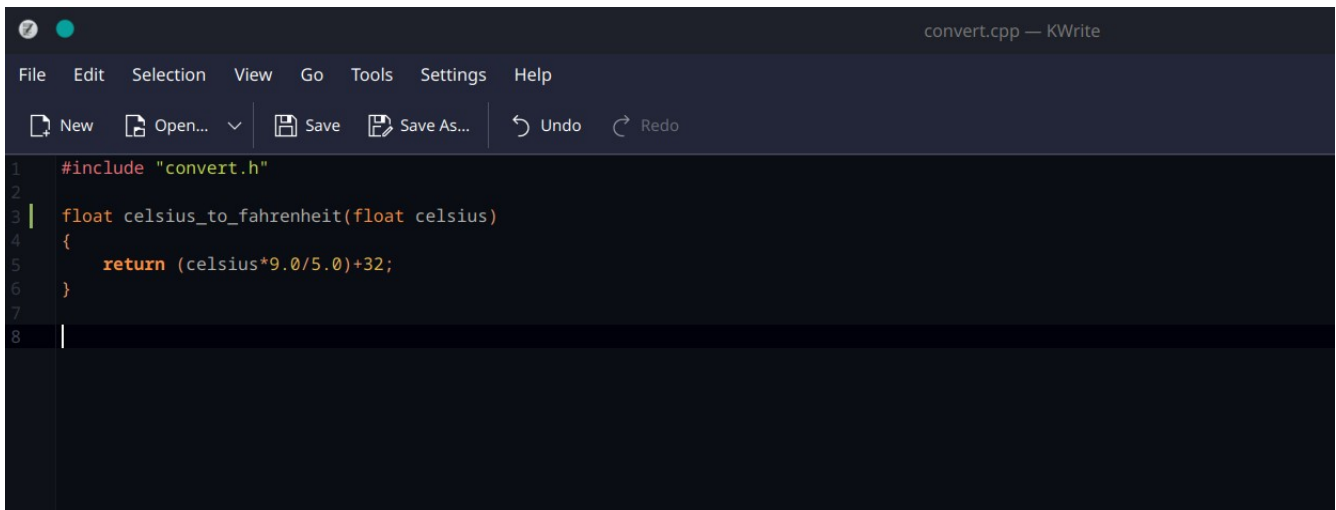
### Code Screenshots:

#### 1.convert.h

A screenshot of the KWrite text editor window titled "convert.h — KWrite". The window has a dark theme. The menu bar includes "File", "Edit", "Selection", "View", "Go", "Tools", "Settings", and "Help". The toolbar shows icons for "New", "Open...", "Save", "Save As...", "Undo", and "Redo". The code area shows the following content:

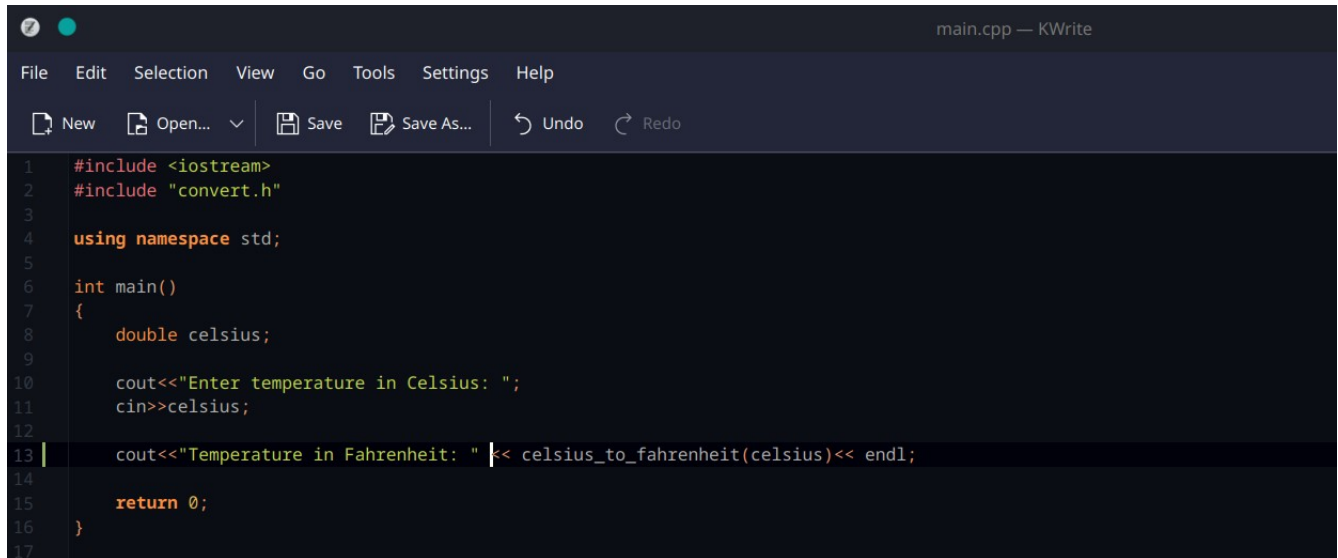
```
1 | float celsius_to_fahrenheit(float celsius);  
2 |  
3 |
```

#### 2.convert.cpp

A screenshot of the KWrite text editor window titled "convert.cpp — KWrite". The window has a dark theme. The menu bar includes "File", "Edit", "Selection", "View", "Go", "Tools", "Settings", and "Help". The toolbar shows icons for "New", "Open...", "Save", "Save As...", "Undo", and "Redo". The code area shows the following content:

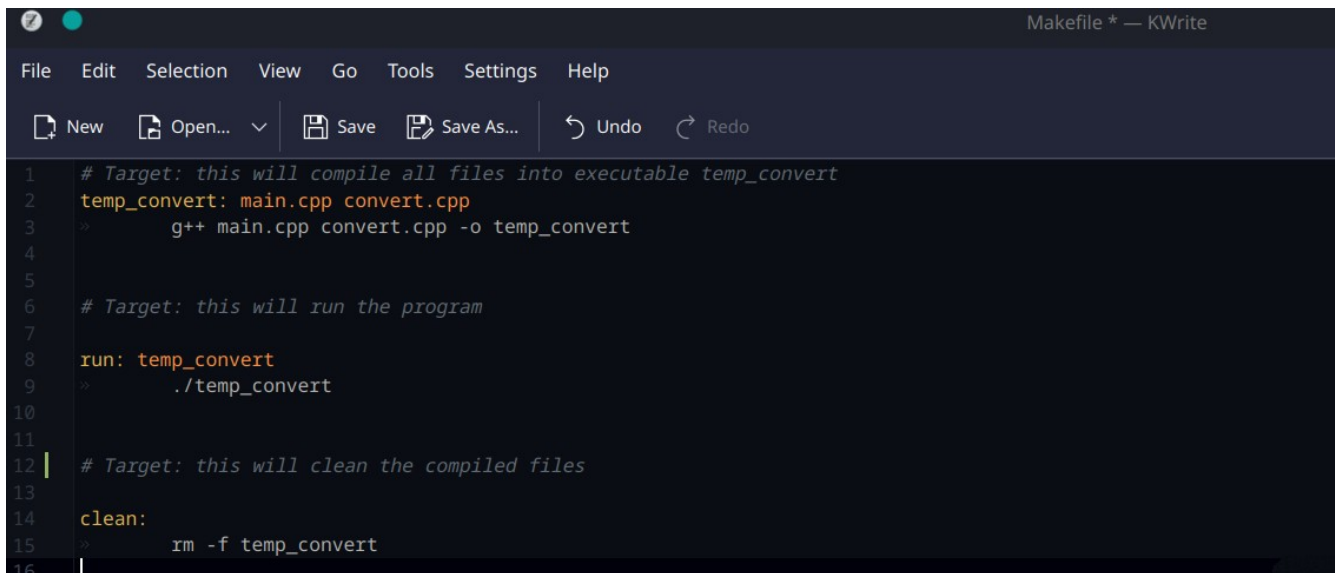
```
1 | #include "convert.h"  
2 |  
3 | float celsius_to_fahrenheit(float celsius)  
4 | {  
5 |     return (celsius*9.0/5.0)+32;  
6 | }  
7 |  
8 |
```

## 3.main.cpp



```
1  #include <iostream>
2  #include "convert.h"
3
4  using namespace std;
5
6  int main()
7  {
8      double celsius;
9
10     cout<<"Enter temperature in Celsius: ";
11     cin>>celsius;
12
13     cout<<"Temperature in Fahrenheit: " << celsius_to_fahrenheit(celsius)<< endl;
14
15     return 0;
16 }
17
```

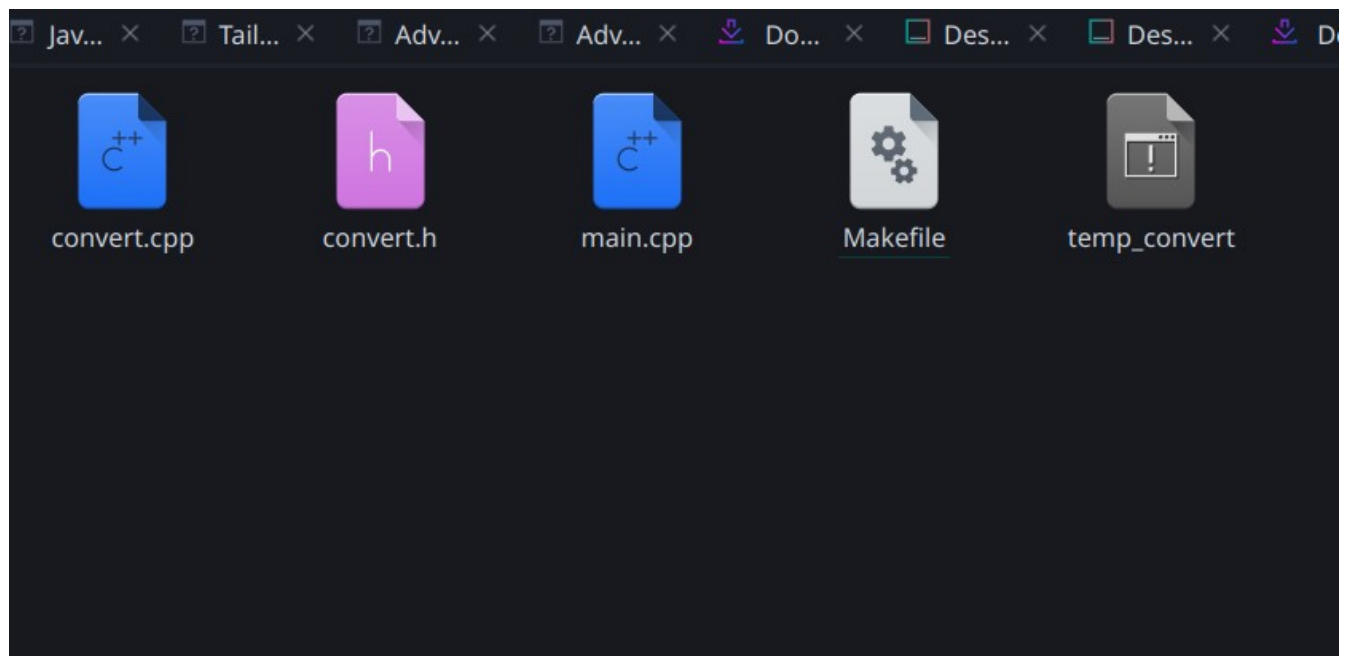
## Make File



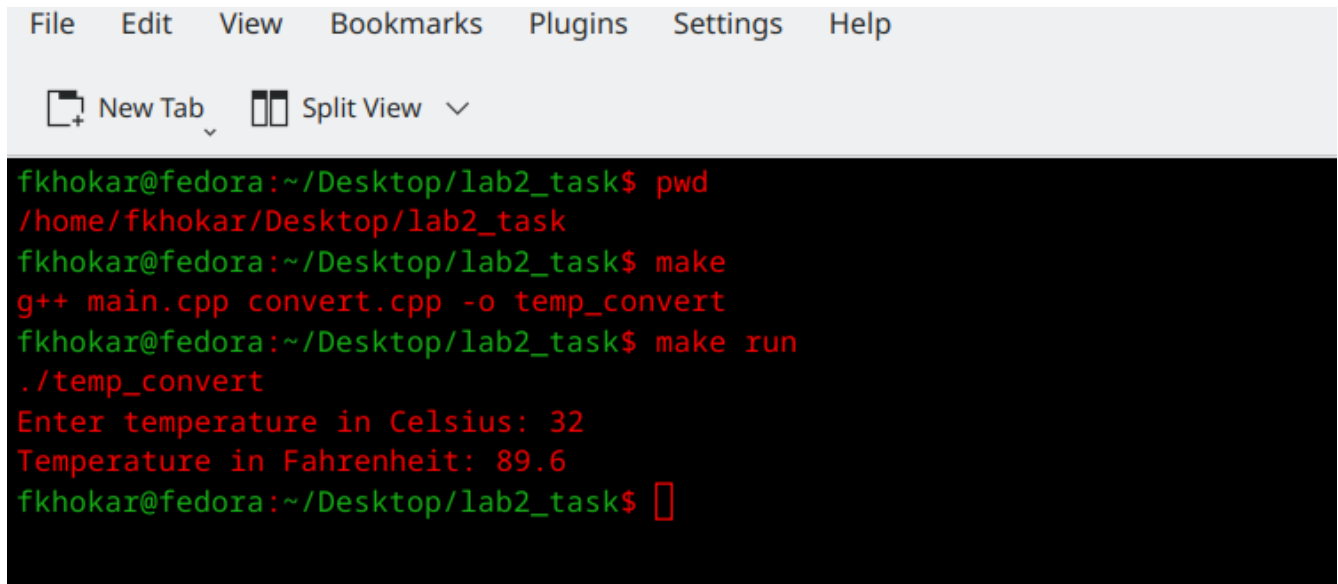
```
1  # Target: this will compile all files into executable temp_convert
2  temp_convert: main.cpp convert.cpp
3  >> g++ main.cpp convert.cpp -o temp_convert
4
5
6  # Target: this will run the program
7
8  run: temp_convert
9  >> ./temp_convert
10
11
12 # Target: this will clean the compiled files
13
14 clean:
15 >> rm -f temp_convert
16
```

## Make -----Command Screenshot:

```
New Tab  Split View  ✓  
fkhokar@fedora:~/Desktop/lab2_task$ pwd  
/home/fkhokar/Desktop/lab2_task  
fkhokar@fedora:~/Desktop/lab2_task$ make  
g++ main.cpp convert.cpp -o temp_convert  
fkhokar@fedora:~/Desktop/lab2_task$
```



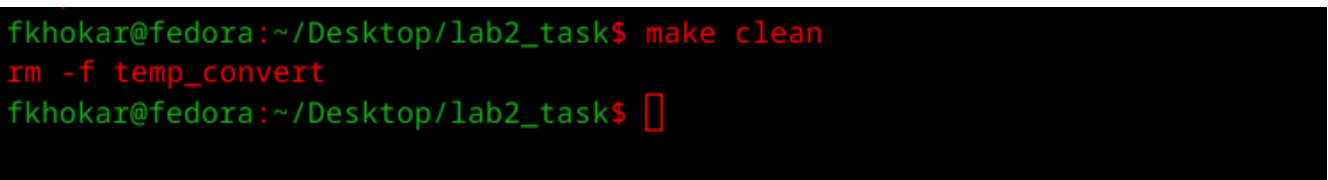
## Make run ----Command Screenshot:



A screenshot of a terminal window with a light gray title bar containing menu items: File, Edit, View, Bookmarks, Plugins, Settings, and Help. Below the title bar, there are icons for 'New Tab' and 'Split View'. The terminal content shows a user named fkhokar at a fedora machine in the directory ~/Desktop/lab2\_task. The user runs 'pwd' and gets the directory path. Then they run 'make', which compiles 'main.cpp' and 'convert.cpp' into 'temp\_convert'. Finally, they run 'make run', which executes './temp\_convert'. The program prompts for a temperature in Celsius (32) and outputs the equivalent in Fahrenheit (89.6). The prompt returns.

```
File Edit View Bookmarks Plugins Settings Help
New Tab Split View
fkhokar@fedora:~/Desktop/lab2_task$ pwd
/home/fkhokar/Desktop/lab2_task
fkhokar@fedora:~/Desktop/lab2_task$ make
g++ main.cpp convert.cpp -o temp_convert
fkhokar@fedora:~/Desktop/lab2_task$ make run
./temp_convert
Enter temperature in Celsius: 32
Temperature in Fahrenheit: 89.6
fkhokar@fedora:~/Desktop/lab2_task$
```

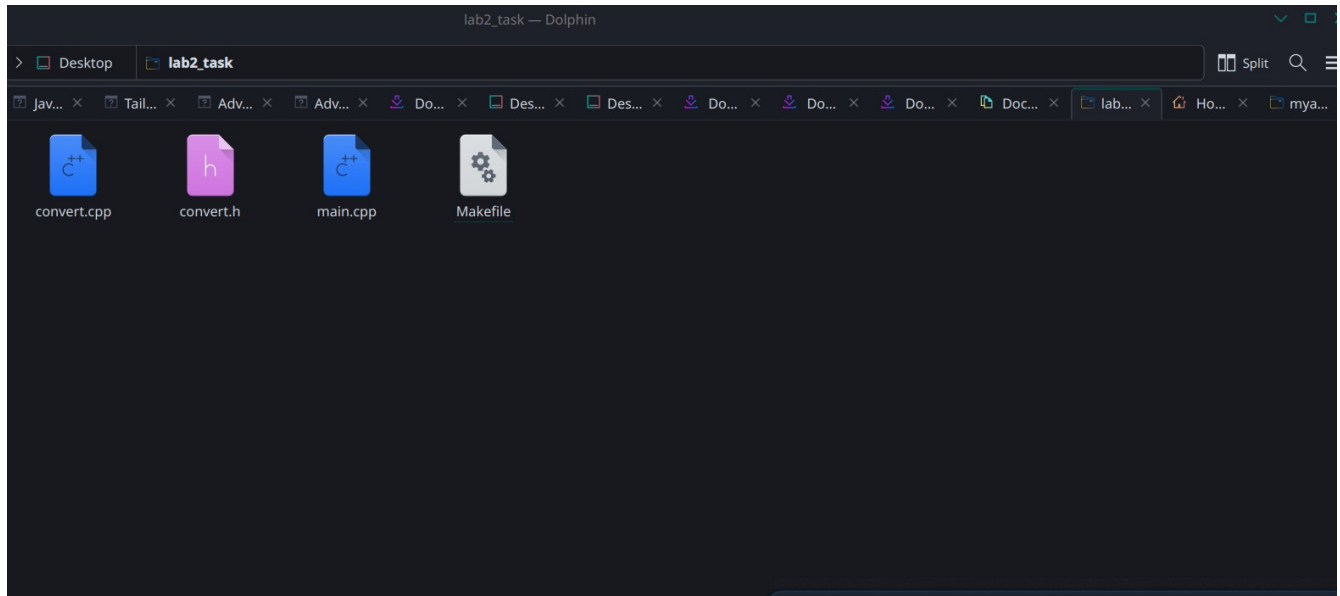
## Make clean---- command screenshot:



A screenshot of a terminal window showing the user fkhokar running 'make clean' in the directory ~/Desktop/lab2\_task. The command successfully removes the 'temp\_convert' file, as indicated by the output 'rm -f temp\_convert'. The prompt then returns.

```
fkhokar@fedora:~/Desktop/lab2_task$ make clean
rm -f temp_convert
fkhokar@fedora:~/Desktop/lab2_task$
```

**Executable file deleted by make clean ---command:**



## Task : 3

### Codes screenshots

#### 1.checkinvalid.h

```
> Makefile x | h checkinvalid.h x
// iin ,h file function declaration
bool is_valid_number(char* string);
int string_to_integer(char* string)
```

## 2.checkinvalid.cpp

```
> % Makefile x h checkinvalid.h x checkinvalid.cpp x main.cpp x
#include "checkinvalid.h"

bool is_valid_number(char* string) // This function will check whether the input string is a valid integer number
{
    int i = 0;

    // this will allow negative number from arguments.....
    if (string[0] == '-')
        i = 1;

    if (string[i] == '\0') // string should not be empty after sign like 123- it should be 123-4
        return false;

    while (string[i] != '\0')
    {
        if (string[i] < '0' || string[i] > '9')
            return false;
        i++;
    }
    return true;
}

int string_to_integer(char* string) // This function will convert a numeric string given through command line in make file into an integer
{
    int i = 0, num = 0, sign = 1;

    if (string[0] == '-')
    {
        sign = -1;
        i = 1;
    }

    while (string[i] != '\0')
    {
        num = num * 10 + (string[i] - '0');
        i++;
    }

    return num * sign;
}
```

## 3.main.cpp

```
> % Makefile x h checkinvalid.h x checkinvalid.cpp x main.cpp x
#include <iostream>
#include "checkinvalid.h"

using namespace std;

int main(int argc, char* argv[]) //argc is for number of inputs and argv is for what value is.....
{
    if (argc < 2)
    {
        cout << "No command line arguments are provided." << endl;
        return 1;
    }

    int arr[argc - 1];

    for (int i = 1; i < argc; i++)
    {
        if (!is_valid_number(argv[i]))
        {
            cout << "Error: Invalid input: " << argv[i] << endl;
            return 1;
        }
        arr[i - 1] = string_to_integer(argv[i]);
    }

    int maximum = arr[0];
    int minimum = arr[0];

    for (int i = 1; i < argc - 1; i++)
    {
        if (arr[i] > maximum)
            maximum = arr[i];
        if (arr[i] < minimum)
            minimum = arr[i];
    }

    cout << "Maximum number is : " << maximum << endl;
    cout << "Minimum number is : " << minimum << endl;

    return 0;
}
```

## Make file screenshot:

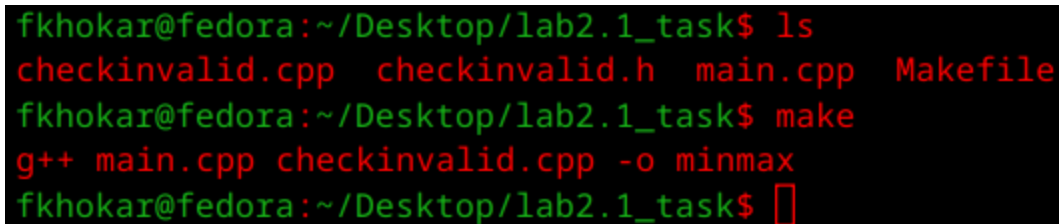


```
> Makefile x checkinvalid.h x checkinvalid.cpp x main.cpp x
# this compile the program into executable 'minmax'
minmax: main.cpp checkinvalid.cpp
>      g++ main.cpp checkinvalid.cpp -o minmax

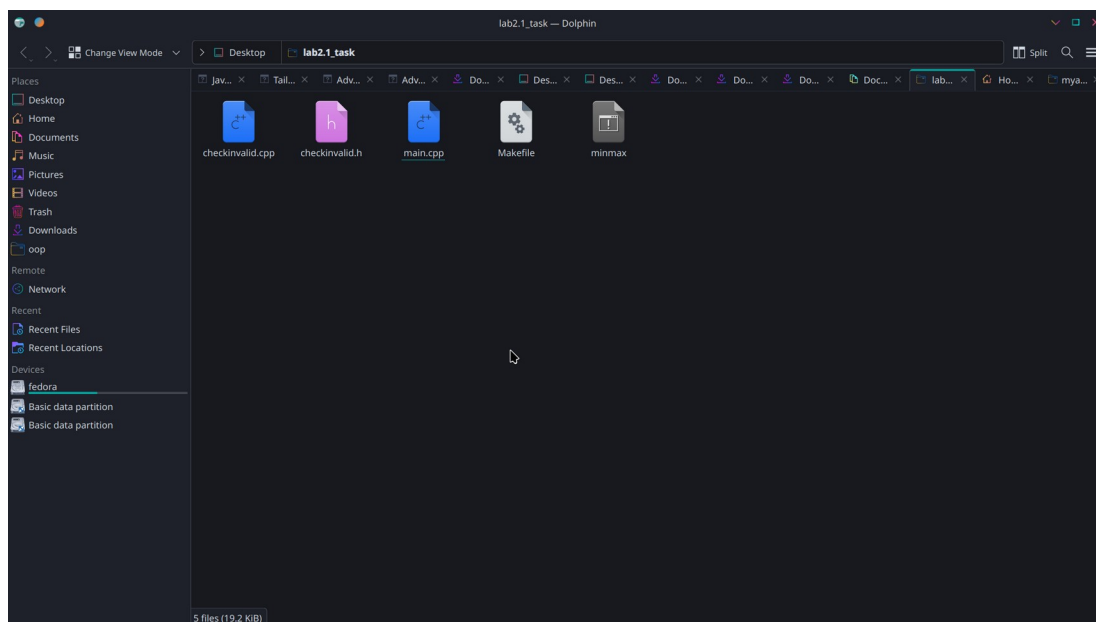
# this will run the program with sample arguments given in this
run: minmax
>      ./minmax 12 6 25 -2 7  #command line argument which will save in array.....

# this will delete compiled files
clean:
>      rm -f minmax
```

## Make ----command screenshot:



```
fkhoar@fedora:~/Desktop/lab2.1_task$ ls
checkinvalid.cpp  checkinvalid.h  main.cpp  Makefile
fkhoar@fedora:~/Desktop/lab2.1_task$ make
g++ main.cpp checkinvalid.cpp -o minmax
fkhoar@fedora:~/Desktop/lab2.1_task$
```





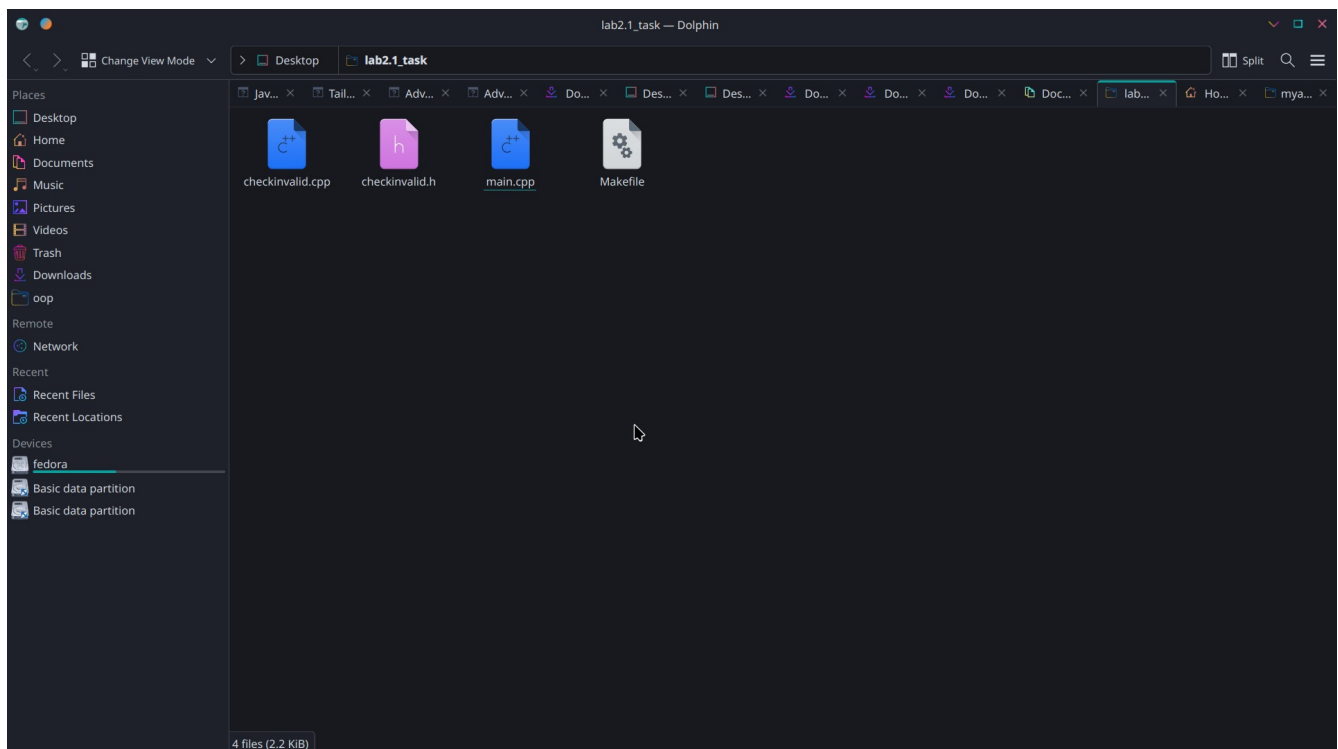
## Make run ----Command screenshot

```
fkhokar@fedora:~/Desktop/lab2.1_task$ make run
./minmax 12 6 25 -2 7      #command line argument which will save in array....
Maximum number is : 25
Minimum number is : -2
fkhokar@fedora:~/Desktop/lab2.1_task$
```

## Make clean -----command sceenshot

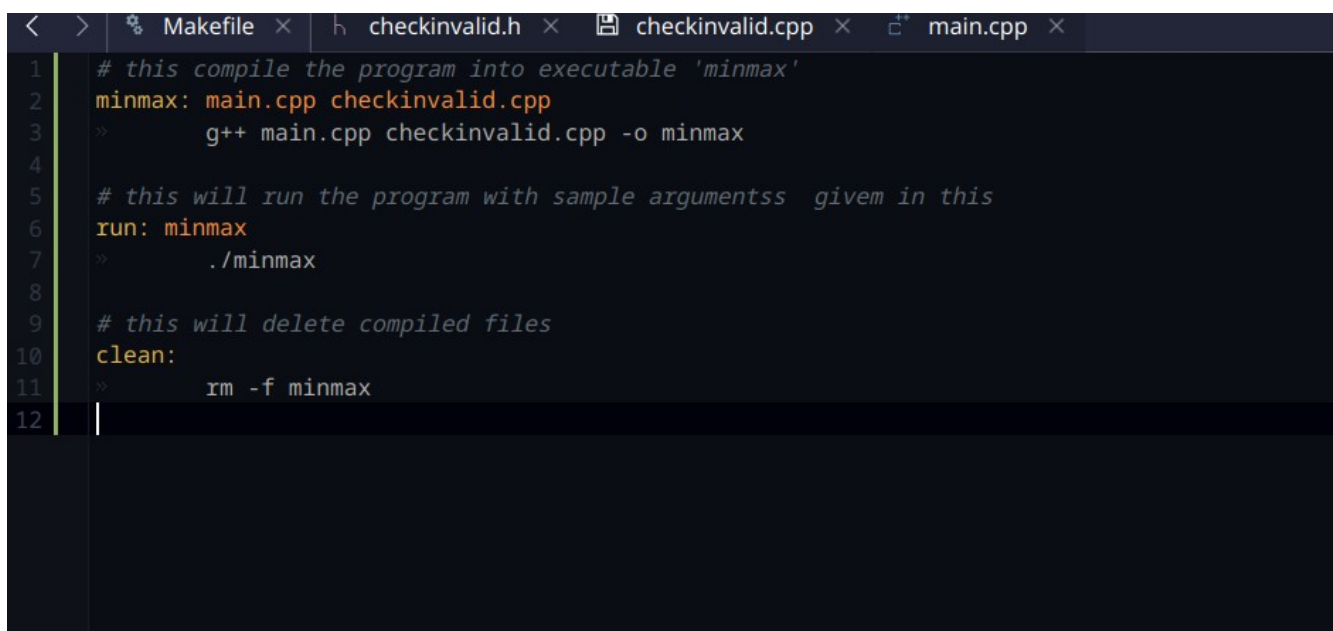
```
fkhokar@fedora:~/Desktop/lab2.1_task$ make clean
rm -f minmax
fkhokar@fedora:~/Desktop/lab2.1_task$
```

## Excutable file is deleted by make clean -----command



Show error when no arguments are passed below is screenshots

```
g++ main.cpp checkinvalid.cpp -o minmax
fkhokar@fedora:~/Desktop/lab2.1_task$ make run
./minmax
No command line arguments are provided.
make: *** [Makefile:7: run] Error 1
fkhokar@fedora:~/Desktop/lab2.1_task$
```



```
< > % Makefile x h checkinvalid.h x checkinvalid.cpp x main.cpp x
1 # this compile the program into executable 'minmax'
2 minmax: main.cpp checkinvalid.cpp
3 >> g++ main.cpp checkinvalid.cpp -o minmax
4
5 # this will run the program with sample argumentss givem in this
6 run: minmax
7 >> ./minmax
8
9 # this will delete compiled files
10 clean:
11 >> rm -f minmax
12
```

Sow error when non numeric value is provided:

```
fkhokar@fedora:~/Desktop/lab2.1_task$ make run
./minmax a b c #command line arrgument which will save in array.....
Error: Invalid input: a
make: *** [Makefile:7: run] Error 1
fkhokar@fedora:~/Desktop/lab2.1_task$
```



```
< > % Makefile x h checkinvalid.h x checkinvalid.cpp x main.cpp x
1 # this compile the program into executable 'minmax'
2 minmax: main.cpp checkinvalid.cpp
3 >> g++ main.cpp checkinvalid.cpp -o minmax
4
5 # this will run the program with sample argumentss givem in this
6 run: minmax
7 >> ./minmax a b c #command line arrgument which will save in array.....
8
9 # this will delete compiled files
10 clean:
11 >> rm -f minmax
12
```